Research, Innovation, Incubation & Commercialization towards Building Successful Startup

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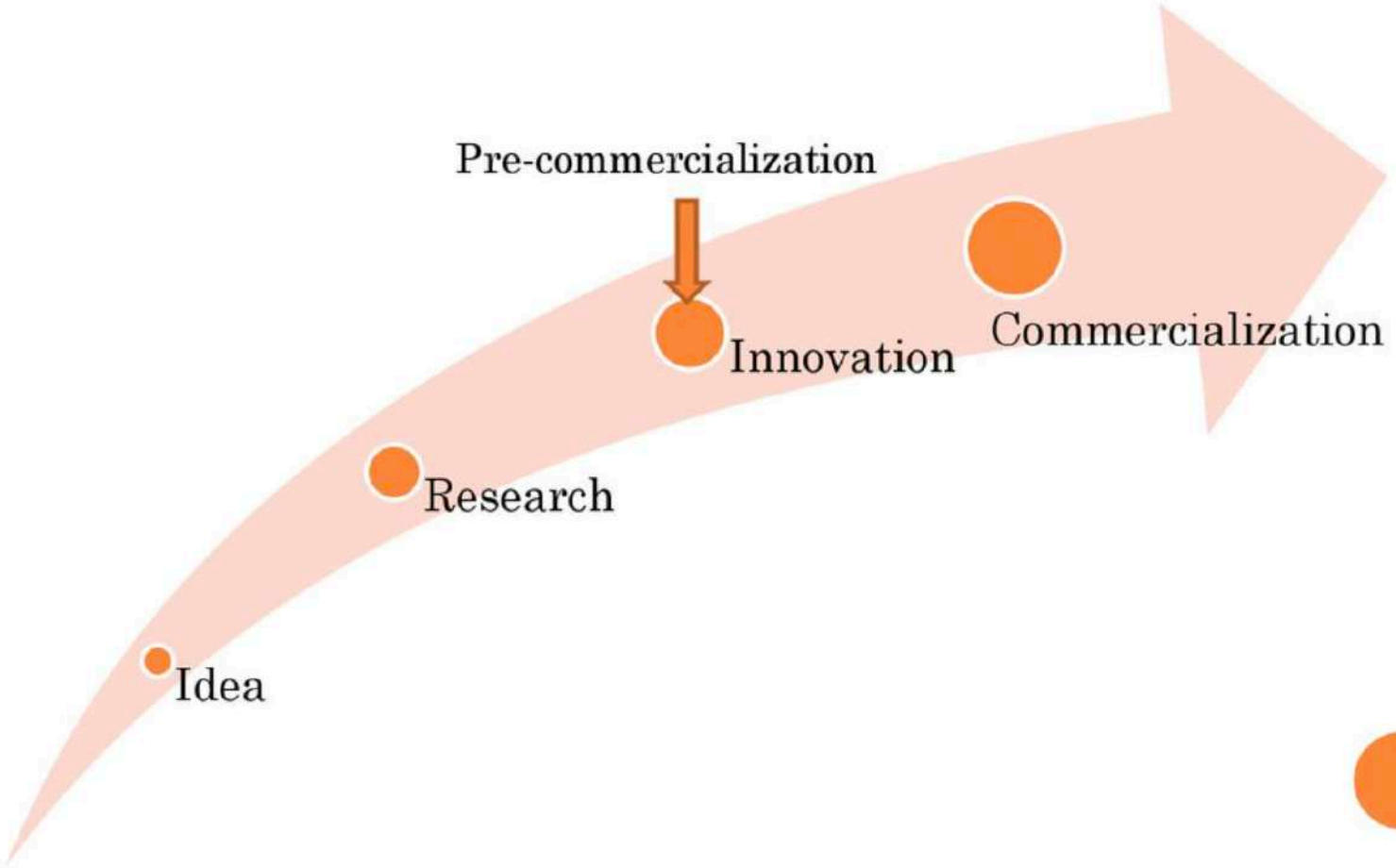
AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

CMED
Connecting People, Saving Lives



Research, Innovation, Incubation, Commercialization

RESEARCH-INNOVATION-COMMERCIALIZATION – GENERAL MODEL

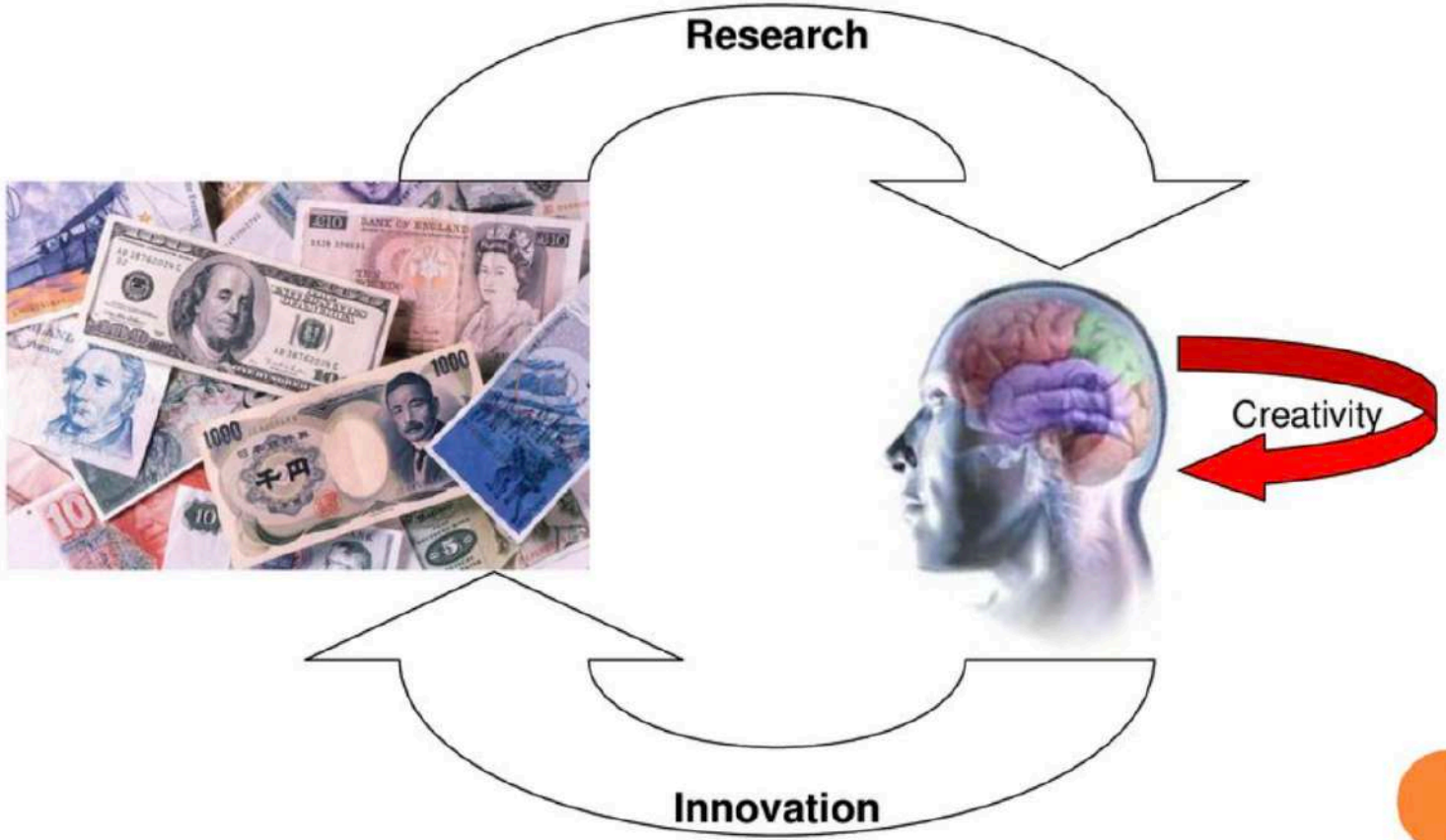



idea

WHAT IS RESEARCH?

- **Research** can be defined as the search for knowledge or as any systematic investigation, with an open mind, to establish novel facts, solve new or existing problems, prove new ideas, or develop new products/services/process

Research vs Innovation – What is the relationship?



Definition of Innovation

Innovation = Invention*Commercialization

The Differences

Innovation

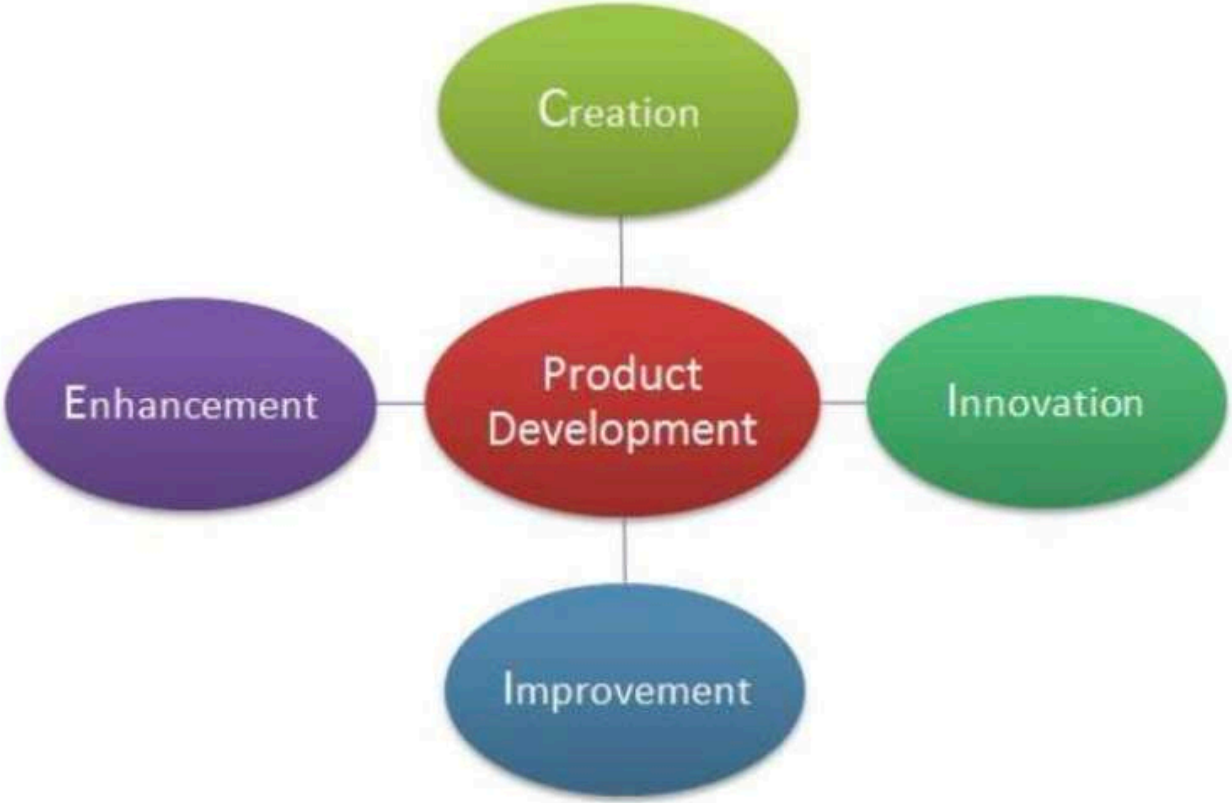
- Innovation happens when someone "improves on or makes a significant contribution" to something that has already been invented. Steve Jobs was an innovator

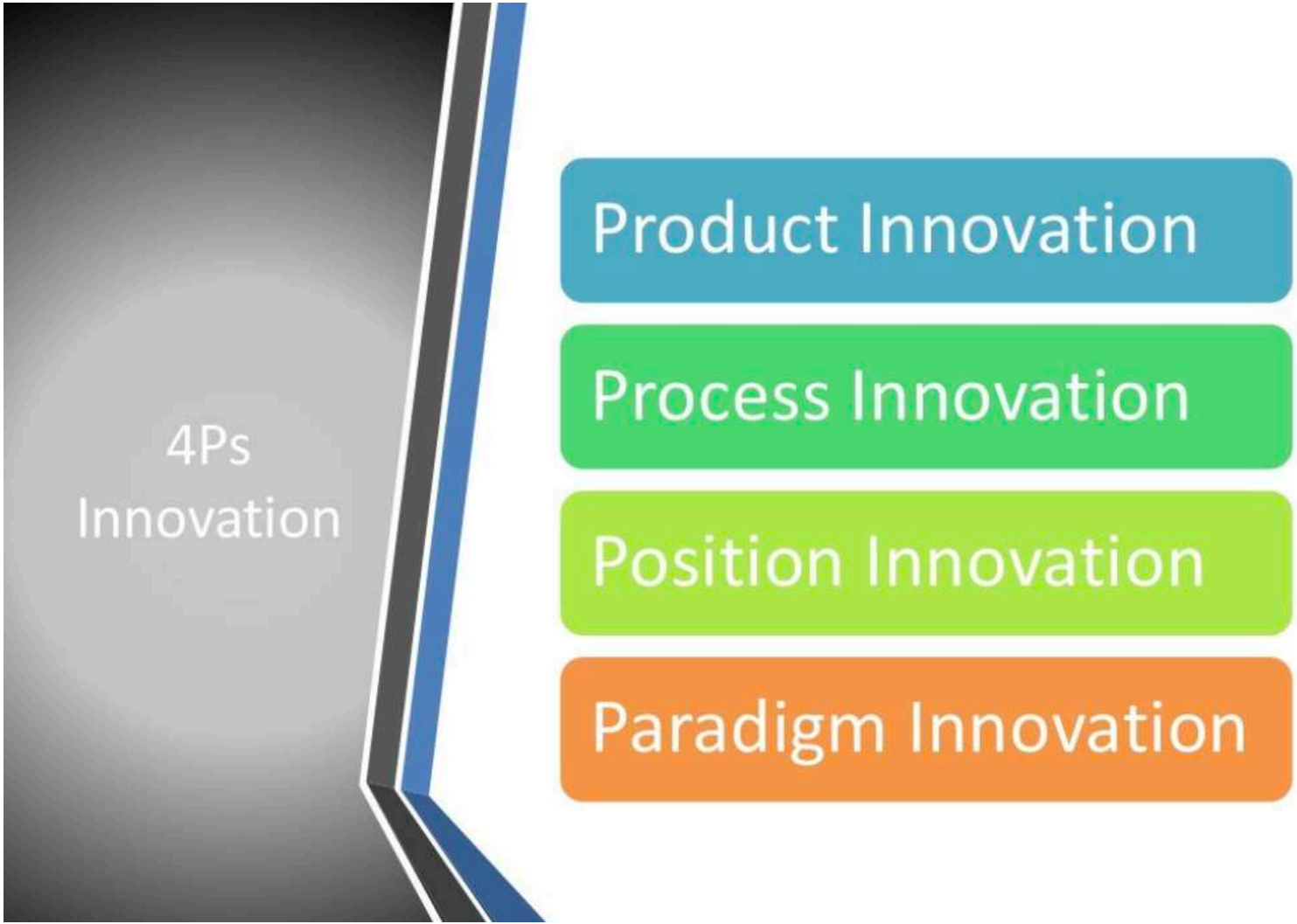
Invention

Invention is the "creation of a product or introduction of a process for the first time." Thomas Edison was an inventor.

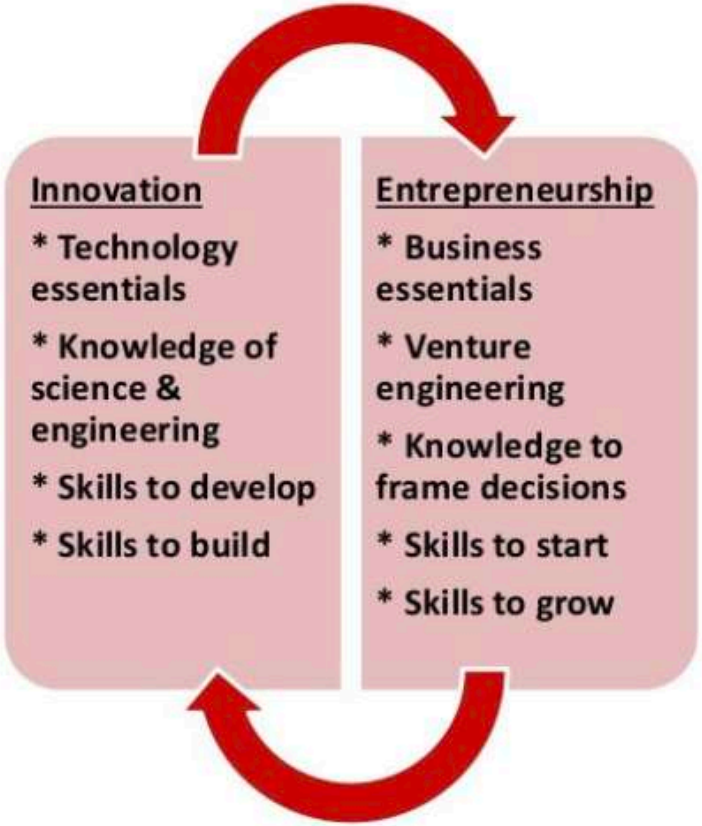
(Bhasin, K., 2012)

Innovation


Creation
Enhancement
Product Development
Innovation
Improvement


4Ps
Innovation
Product Innovation
Process Innovation
Position Innovation
Paradigm Innovation

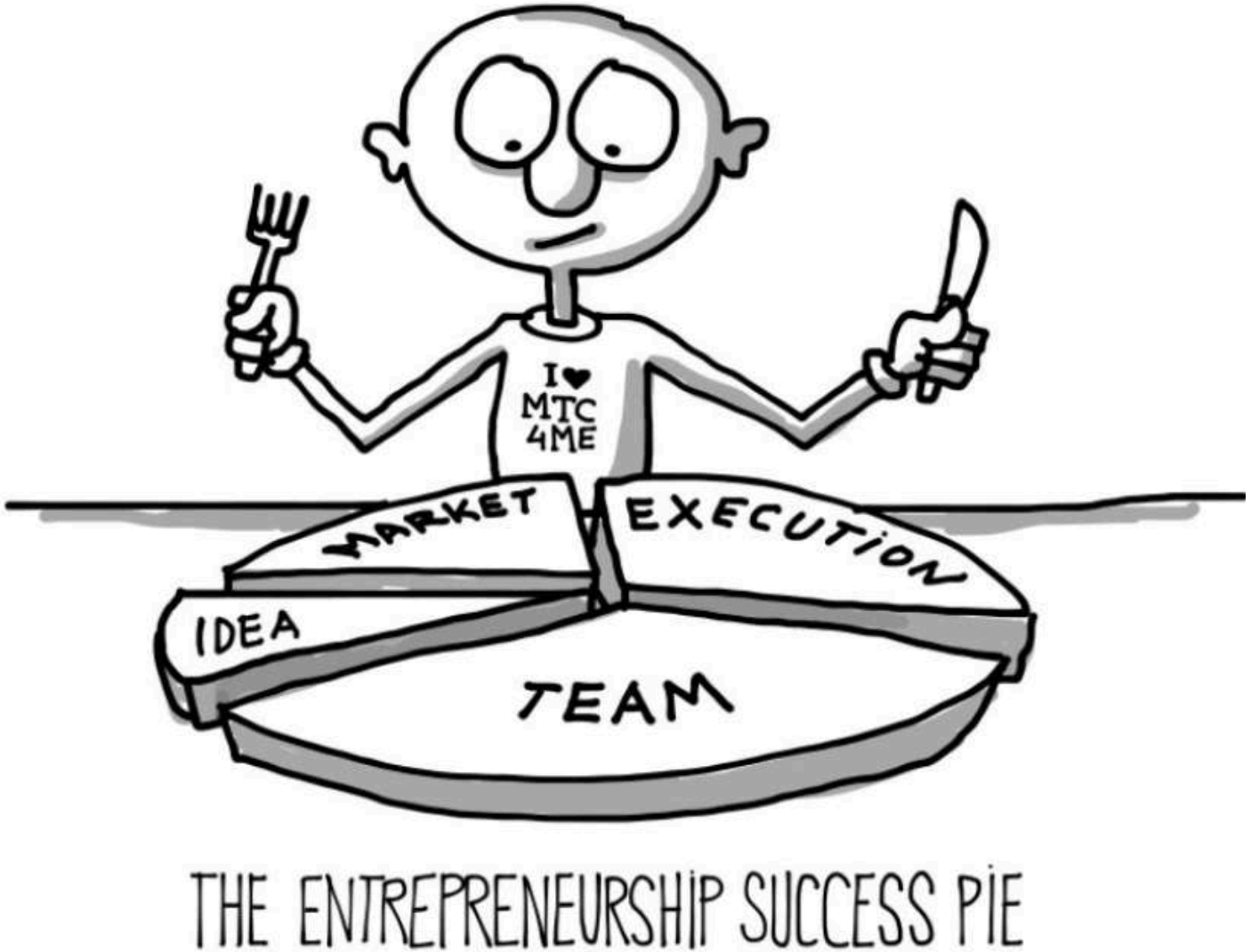
What Is Entrepreneurship?



Successful Entrepreneurship



Spirit + Skills



THE ENTREPRENEURSHIP SUCCESS PIE

New Product Development

- What is the gap in the market?
- What is the trend?
- What is the new product about?
- Who needs the new product?
- How much does it cost?
- Is there enough market to recover the cost?
- Is there any intelligent report as evidence?
- Who are the competitors?



By: Oguchi Martins Egbujor

Intellectual Property Tools

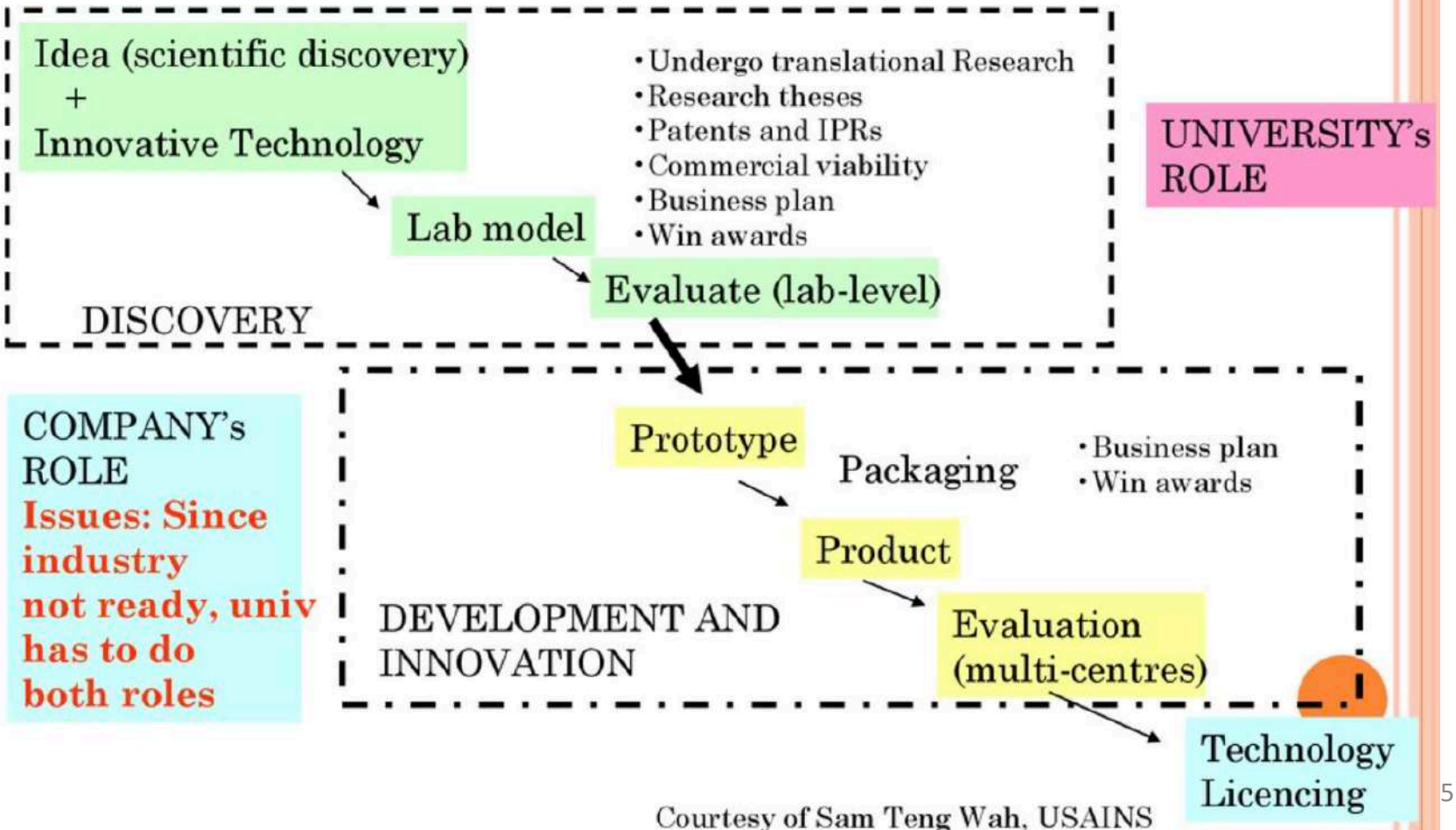
- Copyrights
- Patent
- Trade Mark
- A design

Ownership of Intellectual Property

You own intellectual property if you:

- Created it (and it meets the requirements for copyright, a patent or a design)
- Bought intellectual property rights from the creator or a previous owner
- Have a brand that could be a trade mark, eg a well-known product name
- Intellectual property can:
- have more than one owner
- belong to people or businesses
- be sold or transferred

Commercialization of Knowledge



Courtesy of Sam Teng Wah, USAINS

A startup timeline



Ideation

Potential scalable product/service idea for big enough target market. Some initial revenue models for how it would make money. One person OR only vague team; no confirmed commitment and/or no right skills balance in the team structure yet

Concepting

Having clear and meaningful target with clear direction for min. 3 years with milestones to get there, -> 3, 6, 12, 24, 36 months... Having team of two or three core founding people with balanced ownership. Can also already have some extended team with lighter commitment (stock options and/or cash compensation)

Commitment

Committed & skills balanced founding team. Able to develop the product/service (Minimum Viable Product) without dependency of uncommitted external resources OR already have initial product/ service developed. Have signed shareholder agreement between founders, with milestones, committed time and money usage, for min. 2+ years with vesting etc.

Validation

Can already show some user growth and/or revenue (initial traction). AND/OR continue to attract additional resources (money or sweat equity) for equity or future revenues. Looking for clear market validation (Product Market Fit), to be able to move into scaling.

Scaling

Showing clear, growing and measurable user/market traction in big or rapidly growing target market. Can and want to scale fast. AND/OR is able to attract significant funding.

Establishing

Achieved great growth, that can expected to continue strong. No longer need to "try" get resources and can get those easily. Continue to grow and often wants to culturally continue behaving like a "startup" for as long as possible. Founders make exit or continue biz as usual.

Startup incubators

- **incubator**: Program designed to support successful development of startups by providing business resources.
  - business, marketing, and networking advice
  - computing resources (computers, net access, servers)
  - financial advisors
  - management teams
  - access to Investment, loans and banks
  - access to angel investors and/or venture capitalists
  - legal advisors (such as for intellectual property rights)
  - Examples: GP Accelerator, Ycombinator, Techstars

GP ACCELERATOR

Y Combinator

techstars



https://www.ted.com/talks/bill_gross_the_single_biggest_reason_why_start_ups_succeed?language=en#t-12216